

READING PASSAGE 3

The Physics of Glaciers and Ice Flow

A The Physics of Glaciers and Ice Flow has emerged as one of the most significant topics in contemporary academic discourse. The complexity and multifaceted nature of this subject have attracted the attention of researchers, policymakers, and practitioners across multiple disciplines, all seeking to understand its implications for the future.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D Case studies from diverse geographical and cultural settings provide compelling illustrations of these dynamics in action. In one particularly well-documented example, researchers tracked developments over a fifteen-year period in a setting that offered a natural experiment in the effects of policy change. The introduction of a new regulatory framework produced a cascade of adaptations that rippled through the entire system, generating outcomes that both confirmed theoretical predictions and revealed previously unrecognised complexities.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G In conclusion, the study of the physics of glaciers and ice flow represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 2 The author believes that all research questions in this field have been answered.

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Question 4

Choose the correct letter, A, B, C or D.

- 4 Which of the following best describes the author's view on future developments?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 5

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 5 Case studies have documented the effects of _____ change on the system.

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.

Questions 7–8

Choose the correct letter, A, B, C or D.

- 7 Which of the following best describes the author's view on future developments?
- A. excessive research funding

- B. translating academic findings into practical policy
- C. lack of public interest
- D. too many international collaborations

8 According to the text, what role do structural factors play?

- A. pessimism about the future
- B. cautious optimism with acknowledgment of challenges
- C. complete certainty about outcomes
- D. indifference to practical applications

Question 9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 9** According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Questions 10–11

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 10** The author believes that all research questions in this field have been answered.
- 11** Early researchers had access to the same technologies as modern scholars.

Questions 12–13

Choose the correct letter, A, B, C or D.

12 According to the passage, what is one of the key factors identified by researchers?

- A. weak effects across studies
- B. moderate to large effect sizes with consistency
- C. contradictory findings
- D. results only in specific populations

13 According to the text, what role do structural factors play?

- A. weak effects across studies
- B. moderate to large effect sizes with consistency
- C. contradictory findings
- D. results only in specific populations